

ABHISHEK KULSHRESTHA

Gurugram, India | +91-8279537061 | abhishek.kulshrestha293@gmail.com

[LinkedIn](#) | [GitHub](#) | [Hugging Face](#)

PROFESSIONAL SUMMARY

AI/ML Engineer with hands-on experience building GenAI and agentic AI systems, including real-time voice AI pipelines, RAG-based knowledge assistants, and autonomous tool-calling agents. Skilled in deep learning (CNN, computer vision), LLM orchestration (LangChain, LangGraph), vector search, and agent memory and context management. Experienced designing multi-step, tool-orchestrating agents with guardrails, retry/fallback logic, and LLM evaluation frameworks. Comfortable owning systems end-to-end, from asynchronous FastAPI backends and vector databases to monitoring and latency optimization.

TECHNICAL SKILLS

Languages: Python and GO-lang

Deep Learning / Computer Vision: TensorFlow, PyTorch, Keras, CNN, DNN, Image Classification, Computer Vision, Supervised Learning, Transfer Learning (model fine-tuning)

GenAI / LLMs: LLM application development (GPT-4o, Claude, Gemini, Llama 3/3.2), RAG, LangChain, LangGraph, Prompt Engineering, Fine-Tuning, Sentence Transformers, Model Context Protocol (MCP), Ollama (offline/local LLM deployment)

AI Agents & Orchestration: Tool/API orchestration, multi-step agent workflows, retry and fallback logic, guardrails, short- and long-term agent memory, Pydantic schema validation, LLM-as-a-Judge evaluation

Vector Search & Databases: ChromaDB (vector DB), MongoDB, Redis (caching), PostgreSQL, MySQL

APIs & Backend: FastAPI (async), REST APIs, Flask, Webhooks, Streamlit

Voice AI: LiveKit, WebRTC, Pipecat, Deepgram, ElevenLabs, Whisper, PyAnnote (speaker diarization), Silero VAD

MLOps & Experimentation: MLflow (experiment tracking & model versioning), model evaluation pipelines, performance/latency monitoring

Cloud & DevOps: AWS, Docker, Git/GitHub, Linux (Ubuntu), Nginx, JWT-based auth

Data Analysis: Pandas, NumPy, Matplotlib

QA & Tooling: Manual/Automation/Regression Testing, Selenium, Postman, JIRA, Agile/Scrum

PROFESSIONAL EXPERIENCE

Software Developer (AI / GenAI) — Galactic Client Services

Gurugram, India | Jan 2026 – Present

- **Engineered** asynchronous FastAPI backends for AI agent applications, handling concurrent requests, conversation summarization, structured logging, and exception handling for production workloads.
- **Built** real-time Voice AI pipelines using LivKit, WebRTC, Webhooks, and Pipecat, enabling low-latency STT → LLM → TTS conversation loops and intelligent call transfers.
- **Developed** an AI voice bot for debt-collection/call-center operations, automating customer verification, payment reminders, conversation flows, escalation logic, and live-agent transfers.
- **Integrated** multiple speech and LLM providers — Deepgram, SenseVoice, Whisper, ElevenLabs, Fish Audio, Gemini, and OpenAI — for speech recognition, response generation, and voice synthesis.
- **Optimized** Whisper/PyAnnote speaker diarization pipelines with Silero VAD for speech detection, improving performance by ~26% and reducing speaker-identification error rate from 10% to 7%.
- **Built** an LLM-as-a-Judge evaluation framework to score 1,000+ conversation transcripts against test cases and call-flow criteria, measuring response quality, intent handling, and transfer accuracy.
- **Implemented** Sentence Transformers, PostgreSQL, and Pydantic for semantic search, conversation retrieval, structured LLM outputs, and call analytics.
- **Built** monitoring pipelines to track latency, transcription accuracy, model failures, and voice-bot performance in production.
- **Side build — HR & Attendance Platform:** developed a full-stack Attendance & HR Management System (FastAPI, Streamlit, PostgreSQL) with OTP-based password less JWT authentication, role-based access control for employees/supervisors/admins, and Nginx-proxied deployment on Ubuntu Linux with SSH key-based access.

QA Engineer — Babel Media

Gurugram, Haryana | 2021 – 2022

- Performed manual and functional QA testing across mobile apps, websites, and games, covering gameplay, UI, performance, and localization.
- Logged and tracked defects in JIRA, working with development teams to accelerate resolution.
- Authored test plans and test cases to ensure full feature coverage prior to release.

Career Break — Personal / Medical

2022 – 2025

- Stepped away from full-time work to recover from an accident-related injury. Maintained and grew technical skills through independent study and the self-directed AI/ML projects listed below.

PROJECTS

Jarvis — Real-Time Agentic Voice AI Assistant

Tech: Python, LivKit, Deepgram, ElevenLabs, Local LLMs (Ollama), VAD, Tool Calling, RAG

- **Architected** a real-time multimodal voice assistant using LivKit WebRTC, Deepgram STT, ElevenLabs TTS, and local Ollama (Llama 3/3.2) models, achieving sub-second voice-to-voice latency.
- **Designed** a zero-latency Intent Router (“Fast Path”) that bypassed LLM reasoning for deterministic commands (system queries, media playback, hardware checks), cutting response latency by over 70% (to under 300ms).
- **Implemented** an asynchronous Tool Calling Registry enabling the model to interact with the OS, query web search engines (ddgs), and execute mathematical logic safely direct hands-on experience with goal-based, multi-step agent tool/API orchestration.
- **Built** context-aware screen vision using Tesseract OCR and pygetwindow, allowing the assistant to read, pre-process, and summarize active desktop windows and selected text.
- **Engineered** a hybrid dual-tier agent memory system using Redis for short-term (1-hour) session restoration and MongoDB for long-term user-fact extraction and persistent context injection.

Unreal Engine Blueprint Automation using LLMs and MCP

Tech: Python, LangChain, Pydantic, GPT-4o, Claude, Llama 3, Gemini

- **Built** an agentic tool-calling system that converts natural-language instructions into structured workflows executed through Python functions.
- **Implemented** Pydantic schema validation, multi-step execution, retries, guardrails, and fallback logic to improve agent reliability — directly applicable to safe, explainable multi-agent orchestration.
- **Integrated** multiple LLMs (GPT-4o, Claude, Llama 3, Gemini) for workflow generation and tool execution, comparing outputs across model providers.

RAG-Based Knowledge Assistant Chatbot

Tech: Python, FastAPI, LangChain, LangGraph, ChromaDB, Streamlit

- **Built** a RAG pipeline for document ingestion, semantic search, and context-aware question answering over a ChromaDB vector store.
- **Implemented** hybrid retrieval, chunking, caching, and parallel retrieval, reducing query latency from 2s to 700ms.
- **Developed** asynchronous FastAPI APIs and a Streamlit interface for real-time, document-grounded question answering.

CNN-Based Asset Classification

Tech: Python, TensorFlow, CLIP, OpenCV, FastAPI, ChromaDB, MLflow

- **Built** a multimodal image tagging and retrieval system using CLIP embeddings, cosine similarity, and vector search for classification across unseen categories.
- **Designed** a high-throughput pipeline processing 4K+ images/day, using validation and perceptual hashing to reduce duplicate/noisy data by 30%.
- **Fine-tuned** CLIP-on domain-specific data (transfer learning), improving classification accuracy from 72% to 85%.
- **Developed** asynchronous FastAPI APIs for metadata extraction and batch inference, and integrated MLflow for experiment tracking and model versioning.

Automated Visual Defect Detection System

Tech: Python, TensorFlow, Keras, OpenCV

- **Developed** a CNN-based computer vision system for automated defect detection, achieving an F1-score above 0.90.
- **Built** data augmentation and pre-processing pipelines that increased training data 3x and improved model robustness on unseen images.

Spotify–YouTube Playlist Automation

Tech: Python, Spotify API, YouTube-dl, Pandas

- **Built** an automation tool to retrieve Spotify playlist metadata, search YouTube, validate results, and automate audio downloads, with logging and error handling for reliability.

EDUCATION

Bachelor of Computer Applications — Maharaja Agrasen Himalayan Garhwal University

2020 – 2023

CERTIFICATIONS

- Introduction to Model Context Protocol (MCP)
- AI Fluency Framework & Foundations — Certificate of Completion
- Deep Dive into Deep Learning — Certificate of Excellence
- PyTorch for Deep Learning — Certificate of Excellence